

Example 9.1

We wish to test

$$H_0: \phi_2 = 0 \text{ versus } H_1: \phi_2 \neq 0$$

Assume H_0 is true.

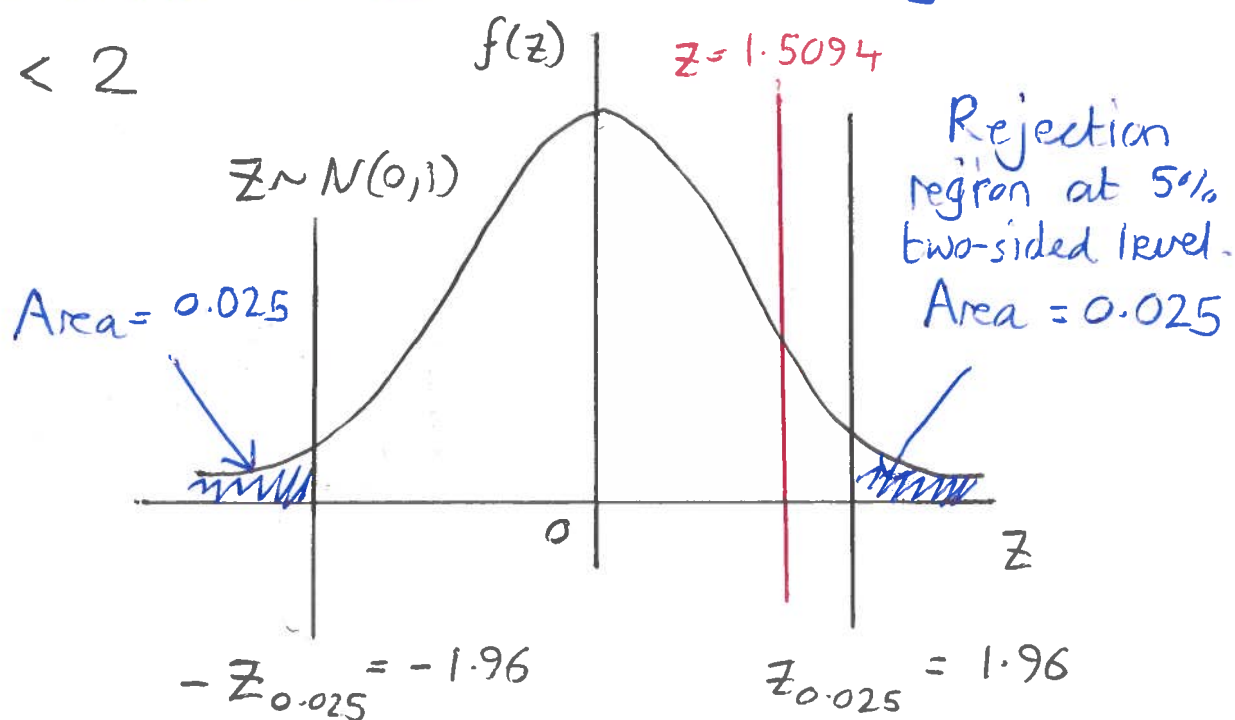
$$\text{Under } H_0: \frac{\hat{\phi}_2}{\text{s.e.}(\hat{\phi}_2)} \sim N(0, 1)$$

Calculate test statistic:

$$\frac{\hat{\phi}_2}{\text{s.e.}(\hat{\phi}_2)} = \frac{0.0477}{0.0316} = 1.5094$$

How 'extreme' is the test statistic?

$$1.5094 < 2$$



For a test at the 5% significance level, $z_{0.025} = 1.96$. We retain / do not reject H_0 and conclude that an AR(1) model is preferable.